

HOMEWORK - 1

1.Sum of two matrices?

```
#include <stdio.h>
int main() {
int a[10][10],b[10][10],c[10][10],n,i,j;
printf("Enter n:");
scanf("%d",&n);
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("Values:");
        scanf("%d",&a[i][j]);
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("Value:");
        scanf("%d",&b[i][j]);
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        c[i][j]=a[i][j]+b[i][j];
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("%d ",c[i][j]);
    }
    printf("\n");
}
return 0;
}
```

OUTPUT:

<pre> #include <stdio.h> int main() { int a[10][10],b[10][10],c[10][10],n,i,j; printf("Enter n:"); scanf("%d",&n); for(i=0;i<n;i++){ for(j=0;j<n;j++){ printf("Values:"); scanf("%d",&a[i][j]); } } for(i=0;i<n;i++){ for(j=0;j<n;j++){ printf("Value:"); scanf("%d",&b[i][j]); } } for(i=0;i<n;i++){ for(j=0;j<n;j++){ c[i][j]=a[i][j]+b[i][j]; } } for(i=0;i<n;i++){ for(j=0;j<n;j++){ printf("%d ",c[i][j]); } printf("\n"); } return 0; } </pre>	<pre> Enter n:2 Values:1 Values:2 Values:3 Values:4 Value:1 Value:2 Value:3 Value:4 2 4 6 8 === Code Execution Successful === </pre>
---	---

2. Write a program in c to read n number of value in an array and display them in reverse order.

```

#include <stdio.h>

int main() {
int a[100],n,i,temp;
printf("Enter n:");
scanf("%d",&n);
for(i=0;i<n;i++){
    scanf("%d",&a[i]);
}

```

```

}
for(i=0;i<n/2;i++){
    temp=a[n-i-1];
    a[n-i-1]=a[i];
    a[i]=temp;
}
for(i=0;i<n;i++){
    printf("%d",a[i]);
}
return 0;
}

```

OUTPUT:

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 int main() { 3 int a[100],n,i,temp; 4 printf("Enter n:"); 5 scanf("%d",&n); 6 for(i=0;i<n;i++){ 7 scanf("%d",&a[i]); 8 } 9 for(i=0;i<n/2;i++){ 10 temp=a[n-i-1]; 11 a[n-i-1]=a[i]; 12 a[i]=temp; 13 } 14 for(i=0;i<n;i++){ 15 printf("%d",a[i]); 16 } 17 return 0; 18 } 19 </pre>	Run Clear <pre> Enter n:5 1 2 3 4 5 54321 === Code Execution Successful === </pre>	

3.Count the total number of duplicate elements in an array.

```

#include <stdio.h>
int main(){
    int a[100],n,i,j,count=0;
    printf("Enter n:");

```

```

scanf("%d",&n);
for(i=0;i<n;i++){
    printf("Values:");
    scanf("%d",&a[i]);
}
for(i=0;i<n;i++){
    for(j=i+1;j<n;j++){
        if(a[i]==a[j]){
            count++;
        }
    }
}
printf("count=%d",count);
return 0;
}

```

OUTPUT:

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 int main(){ 3 int a[100],n,i,j,count=0; 4 printf("Enter n:"); 5 scanf("%d",&n); 6 for(i=0;i<n;i++){ 7 printf("Values:"); 8 scanf("%d",&a[i]); 9 } 10 for(i=0;i<n;i++){ 11 for(j=i+1;j<n;j++){ 12 if(a[i]==a[j]){ 13 count++; 14 } 15 } 16 } 17 printf("count=%d",count); 18 return 0; 19 } 20 21 </pre>		<pre> Enter n:5 Values:1 Values:2 Values:1 Values:2 Values:3 count=2 === Code Execution Successful === </pre>

4.Print unique element in an array.

```
#include <stdio.h>
int main(){
    int a[100],n,i,j,count;
    printf("Enter n:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("Value:");
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++){
        count=0;
        for(j=0;j<n;j++){
            if(a[i]==a[j]){
                count++;
            }
        }
        if(count==1)
            printf("%d",a[i]);
    }
    return 0;
}
```

OUTPUT:

main.c	Output
<pre>1 #include <stdio.h> 2 int main(){ 3 int a[100],n,i,j,count; 4 printf("Enter n:"); 5 scanf("%d",&n); 6 for(i=0;i<n;i++){ 7 printf("Value:"); 8 scanf("%d",&a[i]); 9 } 10 for(i=0;i<n;i++){ 11 count=0; 12 for(j=0;j<n;j++){ 13 if(a[i]==a[j]){ 14 count++; 15 } 16 } 17 if(count==1) 18 printf("%d",a[i]); 19 } 20 return 0; 21 } 22 23</pre>	<pre>Enter n:5 Value:1 Value:2 Value:1 Value:2 Value:3 3 === Code Execution Successful ===</pre>

5.Insert in sorted array.

```
#include <stdio.h>
int main(){
    int a[100],n,i,b;
    printf("Enter the inserting element:");
    scanf("%d",&b);
    printf("Enter n:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("Enter the element:");
        scanf("%d",&a[i]);
    }
    i=n-1;
    while(i>=0 && a[i]>b){
        a[i+1]=a[i];
```

```

        i--;
    }
    a[i+1]=b;
    n++;
    printf("Array:");
    for(i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}

```

OUTPUT:

<pre> 1 #include <stdio.h> 2 int main(){ 3 int a[100],n,i,b; 4 printf("Enter the inserting element :"); 5 scanf("%d",&b); 6 printf("Enter n:"); 7 scanf("%d",&n); 8 for(i=0;i<n;i++){ 9 printf("Enter the element:"); 10 scanf("%d",&a[i]); 11 } 12 i=n-1; 13 while(i>=0 && a[i]>b){ 14 a[i+1]=a[i]; 15 i--; 16 } 17 a[i+1]=b; 18 n++; 19 printf("Array:"); 20 for(i=0;i<n;i++){ 21 printf("%d",a[i]); 22 } 23 return 0; 24 } </pre>	<pre> Enter the inserting element:5 Enter n:5 Enter the element:1 Enter the element:2 Enter the element:3 Enter the element:4 Enter the element:6 Array:123456 === Code Execution Successful === </pre>
---	--

6.Insert in unsorted array.

```
#include <stdio.h>
int main() {
int a[100],n,i,pos,b;
printf("Enter n:");
scanf("%d",&n);
printf("Enter the insert value:");
scanf("%d",&b);
printf("Enter the position:");
scanf("%d",&pos);
for(i=0;i<n;i++){
    printf("Enter the value:");
    scanf("%d",&a[i]);
}
for(i=n-1;i>=pos-1;i--){
    a[i+1]=a[i];
}
a[pos-1]=b;
n++;
for(i=0;i<n;i++){
    printf("%d",a[i]);
}
    return 0;
}
```

OUTPUT:

<pre>1 #include <stdio.h> 2 int main() { 3 int a[100],n,i,pos,b; 4 printf("Enter n:"); 5 scanf("%d",&n); 6 printf("Enter the insert value:"); 7 scanf("%d",&b); 8 printf("Enter the position:"); 9 scanf("%d",&pos); 10 for(i=0;i<n;i++){ 11 printf("Enter the value:"); 12 scanf("%d",&a[i]); 13 } 14 for(i=n-1;i>=pos-1;i--){ 15 a[i+1]=a[i]; 16 } 17 a[pos-1]=b; 18 n++; 19 for(i=0;i<n;i++){ 20 printf("%d",a[i]); 21 } 22 return 0; 23 }</pre>	<pre>Enter n:5 Enter the insert value:5 Enter the position:2 Enter the value:1 Enter the value:2 Enter the value:3 Enter the value:4 Enter the value:5 152345 === Code Execution Successful ===</pre>
---	--

7.Matrix multiplication.

```
#include <stdio.h>

int main() {
    int a[100][100],b[100][100],c[100][100],i,j,n,k;
    printf("Enter n:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        for(j=0;j<n;j++){
            printf("Value:");
            scanf("%d",&a[i][j]);
        }
    }
    for(i=0;i<n;i++){
        for(j=0;j<n;j++){
            printf("Values:");
            scanf("%d",&b[i][j]);
        }
    }
```

```
    }  
}  
for(i=0;i<n;i++){  
    for(j=0;j<n;j++){  
        for(k=0;k<n;k++){  
            c[i][j]+=a[i][k]*b[k][j];  
        }  
    }  
}  
for(i=0;i<n;i++){  
    for(j=0;j<n;j++){  
        printf("%d ",c[i][j]);  
    }  
    printf("\n");  
}  
return 0;  
}
```

OUTPUT:

```

#include <stdio.h>
int main() {
int a[100][100],b[100][100]
    ,c[100][100],i,j,n,k;
printf("Enter n:");
scanf("%d",&n);
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("Value:");
        scanf("%d",&a[i][j]);
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("Values:");
        scanf("%d",&b[i][j]);
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        for(k=0;k<n;k++){
            c[i][j]+=a[i][k]*b[k][j];
        }
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("%d ",c[i][j]);
    }
    printf("\n");
}
return 0;

```

Enter n:2

Value:2

Value:3

Value:4

Value:5

Values:2

Values:3

Values:4

Values:5

16 21

28 37

=== Code Execution Successful ===

8.Matrix transpose.

```

#include <stdio.h>
int main() {
int a[10][10],b[10][10],n,i,j;
printf("Enter n:");
scanf("%d",&n);
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("Enter the value:");

```

```
        scanf("%d",&a[i][j]);
    }
}
for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        b[j][i]=a[i][j];
    }
}

for(i=0;i<n;i++){
    for(j=0;j<n;j++){
        printf("%d ",b[i][j]);
    }
    printf("\n");
}
return 0;
}
```

OUTPUT:

<pre>#include <stdio.h> int main() { int a[10][10], b[10][10], n, i, j; printf("Enter n:"); scanf("%d", &n); for(i=0; i<n; i++){ for(j=0; j<n; j++){ printf("Enter the value:"); scanf("%d", &a[i][j]); } } for(i=0; i<n; i++){ for(j=0; j<n; j++){ b[j][i] = a[i][j]; } } for(i=0; i<n; i++){ for(j=0; j<n; j++){ printf("%d ", b[i][j]); } printf("\n"); } return 0; }</pre>	<pre>Enter n:2 Enter the value:1 Enter the value:2 Enter the value:3 Enter the value:4 1 3 2 4 === Code Execution Successful ===</pre>
---	---

9.Right diagonal sum.

```
#include <stdio.h>
int main() {
    int a[100][100], i, j, n, sum=0;
    printf("Enter n:");
    scanf("%d", &n);
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            printf("Value:");
            scanf("%d", &a[i][j]);
        }
    }
    for(i=0; i<n; i++){
        sum+=a[i][i];
    }
```

```

}
printf("Right diagonal sum:%d",sum);
return 0;
}

```

OUTPUT:

<pre> #include <stdio.h> int main() { int a[100][100],i,j,n,sum=0; printf("Enter n:"); scanf("%d",&n); for(i=0;i<n;i++){ for(j=0;j<n;j++){ printf("Value:"); scanf("%d",&a[i][j]); } } for(i=0;i<n;i++){ sum+=a[i][i]; } printf("Right diagonal sum:%d",sum); return 0; } </pre>	<pre> Enter n:2 Value:1 Value:2 Value:3 Value:4 Right diagonal sum:5 === Code Execution Successful === </pre>
---	--

10.Check for identity matrix.

```

#include <stdio.h>
int main() {
    int a[100][100],i,j,n,m,b=1;
    printf("Enter n:");
    scanf("%d",&n);
    printf("Enter m:");
    scanf("%d",&m);
    for(i=0;i<n;i++){
        for(j=0;j<m;j++){
            printf("Value:");
            scanf("%d",&a[i][j]);
        }
    }
}

```

```

}
if(n==m){
for(i=0;i<n;i++){
    for(j=0;j<m;j++){
        if((i==j && a[i][j]!=1) || (i!=j && a[i][j]!=0)){
            b=0;
            break;
        }
    }
}
if(b==0){
    break;
}
}
if(b==1){
    printf("Identity matrix");
}
else{
    printf("Not a identity matrix");
}
    return 0;
}

```

OUTPUT:

```
printf("Enter m:");
scanf("%d",&m);
for(i=0;i<n;i++){
    for(j=0;j<m;j++){
        printf("Value:");
        scanf("%d",&a[i][j]);
    }
}
if(n==m){
for(i=0;i<n;i++){
    for(j=0;j<m;j++){
        if((i==j && a[i][j]!=1)||(i!=j
        && a[i][j]!=0 )){
            b=0;
            break;
        }
    }
    if(b==0){
        break;
    }
}
```

```
Enter n:3
Enter m:3
Value:1
Value:0
Value:0
Value:0
Value:1
Value:0
Value:0
Value:0
Value:1
Identity matrix

=== Code Execution Successful ===
```

11.To find number of vowels, consonants, digits and white spaces.

```
#include <stdio.h>
int main() {
int i,n,vowel=0,cons=0,words=0;
char a[100];
printf("Enter the no of characters:");
scanf("%d",&n);
scanf("%s",a);
for(i=0;i<n;i++){
    if(a[i]=='a' || a[i]=='i' || a[i]=='o' || a[i]=='e' || a[i]=='u'){
        vowel++;
    }
    if(a[i]==' '){
        words++;
    }
    if((a[i]<='z'&& a[i]>'a')&&(a[i]!='i'&&a[i]!='o'&&a[i]!='e'&&a[i]!='u')){
        cons++;
    }
}
```



```

}
printf("Vowels:%d\nConsonants:%d\nWords:%d",vowel,cons,words+1);
return 0;
}

```

OUTPUT:

main.c	Output
<pre> 1 #include <stdio.h> 2 int main() { 3 int i,n,vowel=0,cons=0,words=0; 4 char a[100]; 5 printf("Enter the no of characters:"); 6 scanf("%d",&n); 7 scanf("%s",a); 8 for(i=0;i<n;i++){ 9 if(a[i]=='a' a[i]=='i' a[i] == 'o' a[i]=='e' a[i]=='u'){ 10 vowel++; 11 } 12 if(a[i]==' '){ 13 words++; 14 } 15 if((a[i]<='z' && a[i]>'a') && (a[i]! ='i' && a[i]!='o' && a[i]! ='e' && a[i]!='u')){ 16 cons++; 17 } 18 } 19 printf("Vowels:%d\nConsonants:%d\nWords :%d",vowel,cons,words+1); 20 return 0; 21 } 22 </pre>	<pre> Enter the no of characters:11 vivekashree Vowels:5 Consonants:6 Words:1 === Code Execution Successful === </pre>

12. Frequency of a character in a string.

```

#include <stdio.h>
#include<string.h>
int main() {
char a[100],temp[100];
int i,j=0,count=0;

```

```

printf("Enter the string:");
scanf("%99[^\n]",a);
printf("Enter the word:");
scanf("%s",temp);
for(i=0;a[i]!='\0';i++){
j=0;
while(temp[j]!='\0' && a[i+j] == temp[j]){
j++;
}
if(word[j]=='\0'){
count++;
}
}
printf("Frequency:",count);
return 0;
}

```

OUTPUT:

```

int main()
{
    char str[100], word[50];
    int i, j, count = 0;

    printf("Enter the string: ");
    scanf("%99[^\n]", str);

```

Enter the string: this is new world and it is new for me and I love it
Enter the word: it
Frequency = 2

=== Code Execution Successful ===

13.Count words in a string.

```

#include <stdio.h>
#include<string.h>
int main() {
char a[100];
int i,j=0;
printf("Enter a string:");
scanf("%49[^\n]",a);
for(i=0;i<strlen(a);i++){
if(a[i]==' '){

```

```

        j++;
    }
}
printf("Words:%d",j+1);
return 0;
}

```

main.c	Output
<pre> 1 #include <stdio.h> 2 #include<string.h> 3 int main() { 4 char a[100]; 5 int i,j=0; 6 printf("Enter a string:"); 7 scanf("%49[^\n]",a); 8 for(i=0;i<strlen(a);i++){ 9 if(a[i]==' '){ 10 j++; 11 } 12 } 13 printf("Words:%d",j+1); 14 return 0; 15 } 16 </pre>	<pre> Enter a string:this is my new world Words:5 === Code Execution Successful === </pre>

OUTPUT:

14. Sort string array.

```

#include <stdio.h>
#include<string.h>
int main() {
char a[100][50],temp[100];
int i,j=0,n;
printf("Enter n:");
scanf("%d",&n);
for(i=0;i<n;i++){
scanf("%s",a[i]);
}
for(i=0;i<n-1;i++){
for(j=0;j<n-i-1;j++){
if (strcmp(a[j],a[j+1])>0){

```

```

        strcpy(temp,a[j]);
        strcpy(a[j],a[j+1]);
        strcpy(a[j+1],temp);
    }
}
}
for(i=0;i<n;i++){
    printf("%s \n",a[i]);
}
return 0;
}

```

OUTPUT:

<pre> 1 #include <stdio.h> 2 #include<string.h> 3 int main() { 4 char a[100][50],temp[100]; 5 int i,j=0,n; 6 printf("Enter n:"); 7 scanf("%d",&n); 8 for(i=0;i<n;i++){ 9 scanf("%s",a[i]); 10 } 11 for(i=0;i<n-1;i++){ 12 for(j=0;j<n-i-1;j++){ 13 if (strcmp(a[j],a[j+1])>0){ 14 strcpy(temp,a[j]); 15 strcpy(a[j],a[j+1]); 16 strcpy(a[j+1],temp); 17 } 18 } 19 } 20 for(i=0;i<n;i++){ 21 printf("%s \n",a[i]); 22 } 23 return 0; 24 } </pre>	<pre> Enter n:4 viveka saravanan kumutha naresh kumutha naresh saravanan viveka === Code Execution Successful === </pre>
---	---

15.Toggle case of a sentence(Lowercase to Uppercase or vice versa)

```
#include <stdio.h>

int main() {
    char a[100];
    int i,n;
    printf("Enter n:");
    scanf("%d",&n);
    printf("Enter the string:");
    scanf("%s",a);
    for(i=0;i<n;i++){
        if(a[i]>='a' && a[i]<='z'){
            a[i]=a[i]-32;
        }
        else if(a[i]>='A' && a[i]<='Z'){
            a[i]=a[i]+32;
        }
    }
    printf("String:%s",a);
    return 0;
}
```

OUTPUT:

main.c	Output
<pre>1 2 #include <stdio.h> 3 int main() { 4 char a[100]; 5 int i,n; 6 printf("Enter n:"); 7 scanf("%d",&n); 8 printf("Enter the string:"); 9 scanf("%s",a); 10 for(i=0;i<n;i++){ 11 if(a[i]>='a' && a[i]<='z'){ 12 a[i]=a[i]-32; 13 } 14 else if(a[i]>='A' && a[i]<='Z'){ 15 a[i]=a[i]+32; 16 } 17 } 18 printf("String:%s",a); 19 return 0; 20 }</pre>	<pre>Enter n:11 Enter the string:VIVEKASHREE String:vivekashree === Code Execution Successful ===</pre>